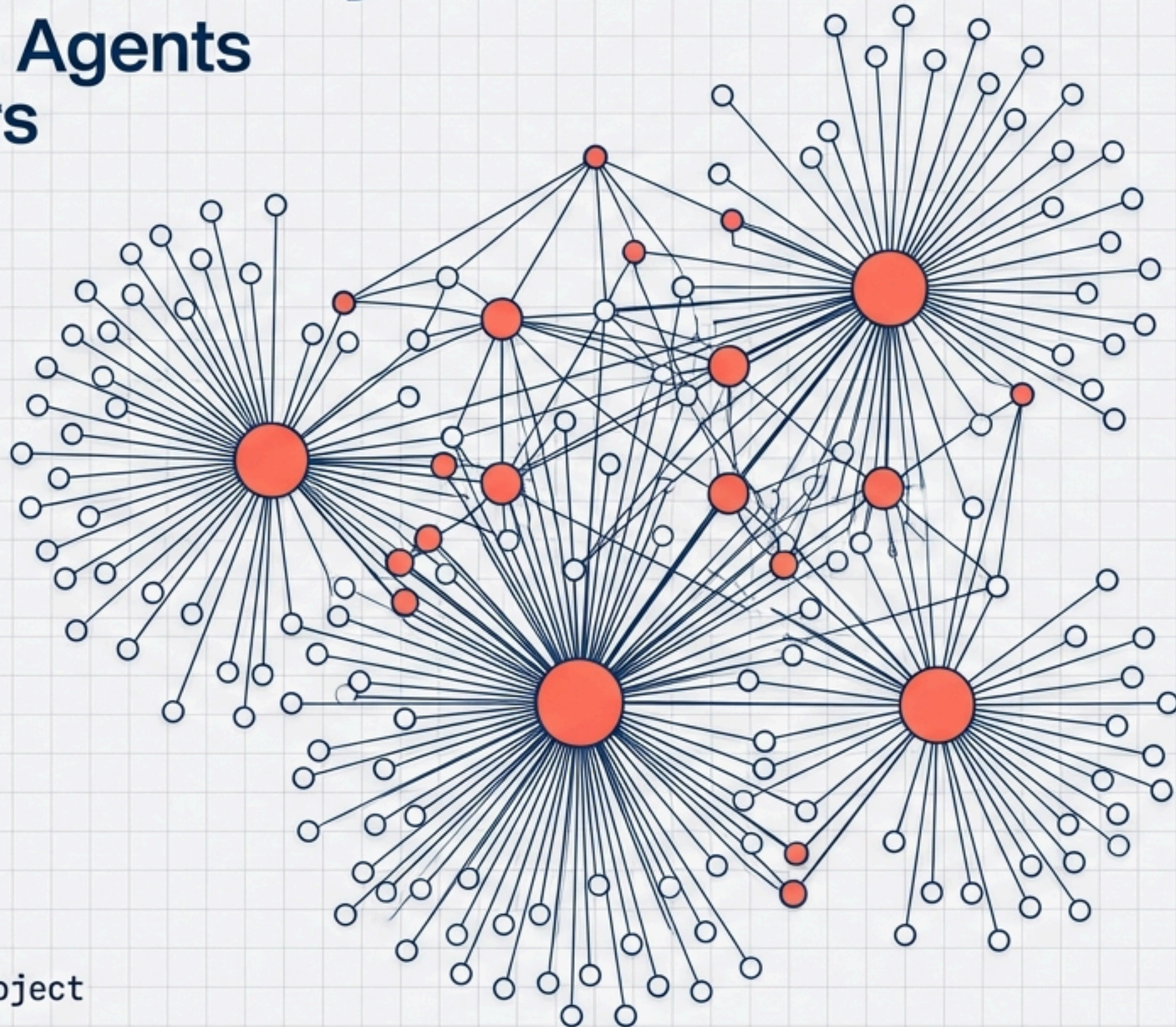
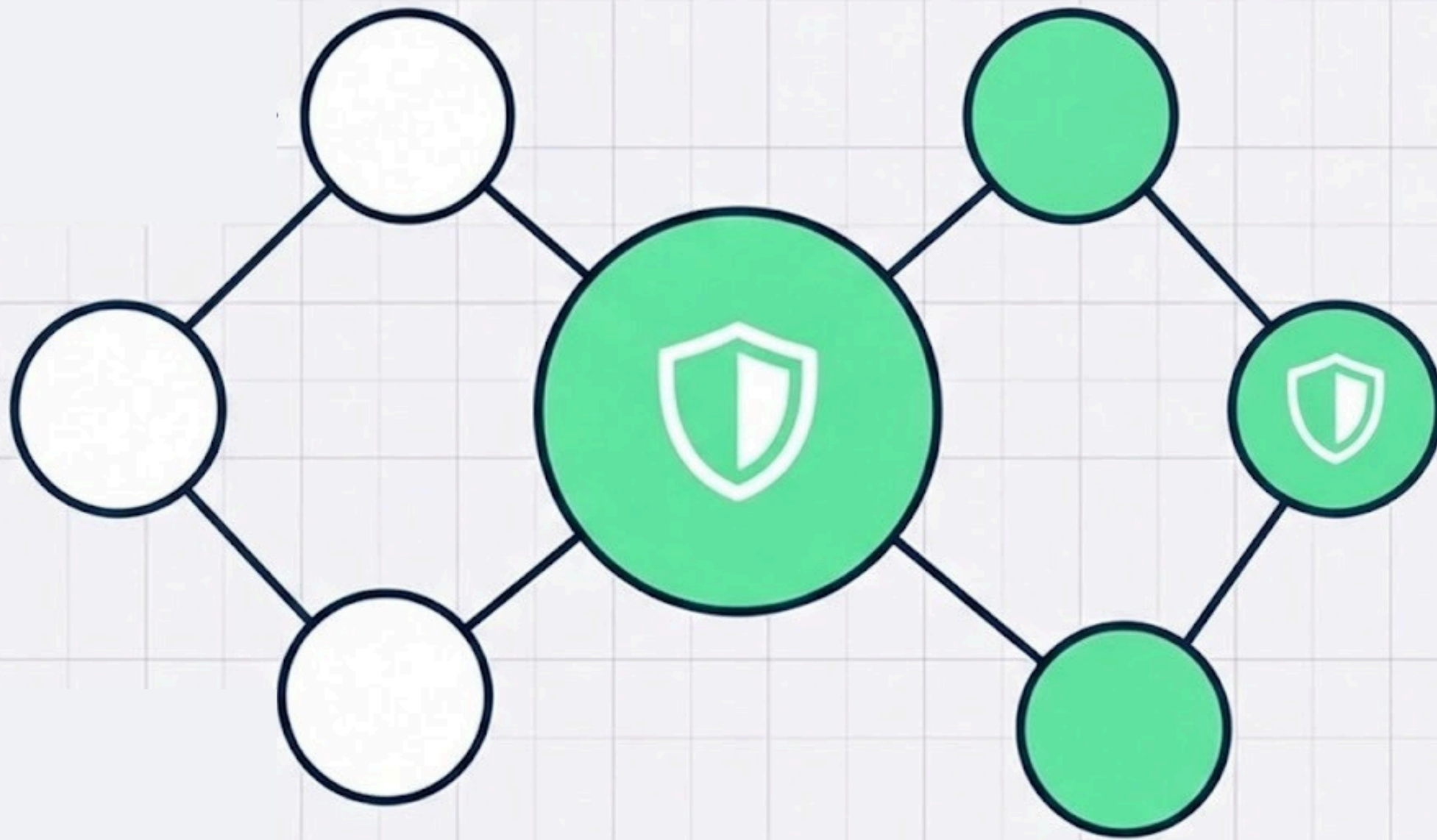


Network Security Games

From Strategic Agents
to Logic Solvers



The Objective



Goal: Secure Graph $G=(V,E)$

The Rule: A set S is valid if every node is in S OR has a neighbor in S .

Equivalence: Minimal Vertex Cover (NP-Hard)

Task 1: Strategic Game Modeling

Investing Node (Secure)



Net Payoff = 6

Free-riding Node (Covered)



Net Payoff = 10

Game Type: Strategic Substitutes (Local Public Good)

Parameters: Value $\alpha=10$, Cost $c=4$ | **Goal:** Find Pure Nash Equilibria (PNE)

Reaching Equilibrium



Best Response Dynamics

Strictly improving updates.

Sequential logic prevents cycles.



Fictitious Play

History-based beliefs.

Deterministic threshold response.

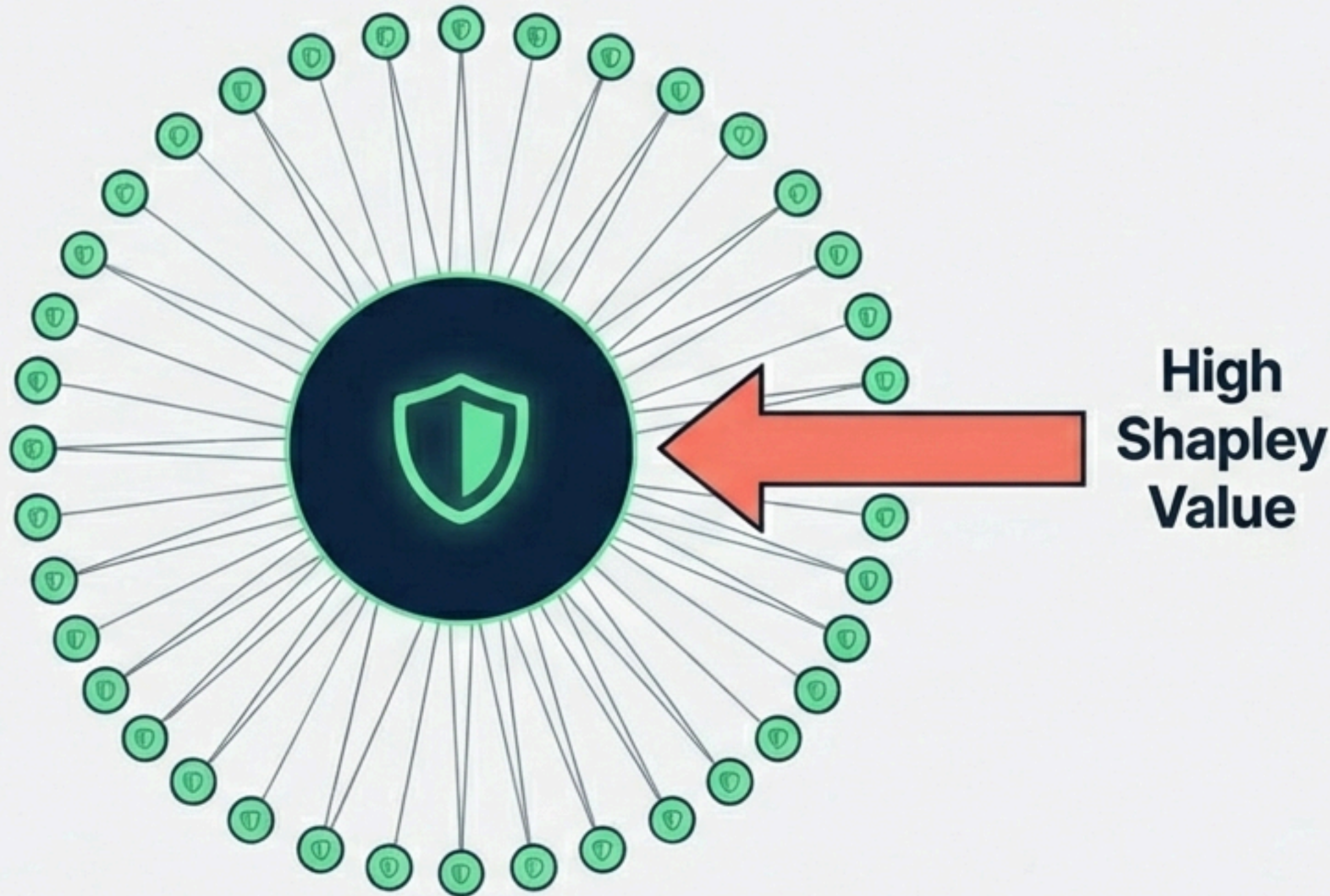


Regret Matching

Minimizes cumulative regret.

Probabilistic selection.

Task 2: Coalitional Analysis



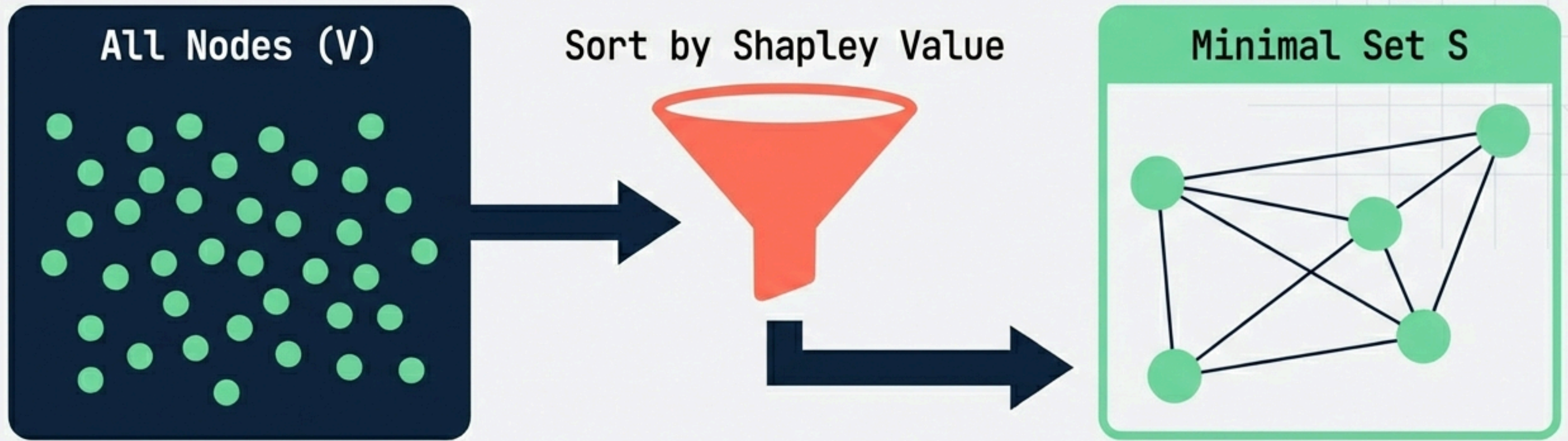
Metric: Shapley Value
(Average Marginal Contribution)

Challenge: $N!$ Complexity
(Intractable)

Solution: Monte Carlo Approximation

Contribution = +1 if
node is the **last**
missing neighbor.

The Reverse Greedy Heuristic

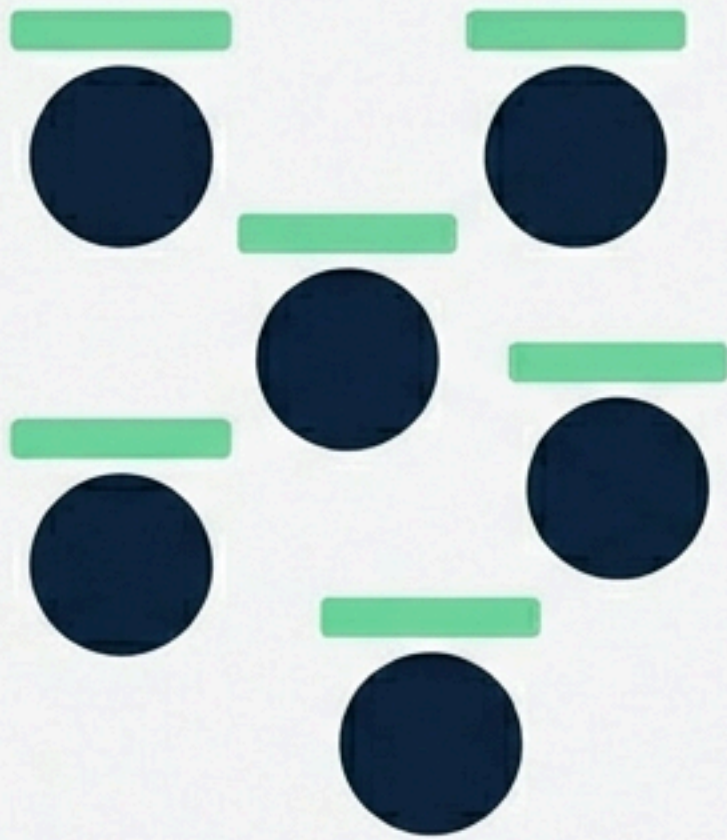


Logic: Start Full -> Prune Sequentially

Constraint: Remove node 'i' ONLY IF neighbors remain safe.

Task 3: Market Allocation

Secure Nodes (Budget B)



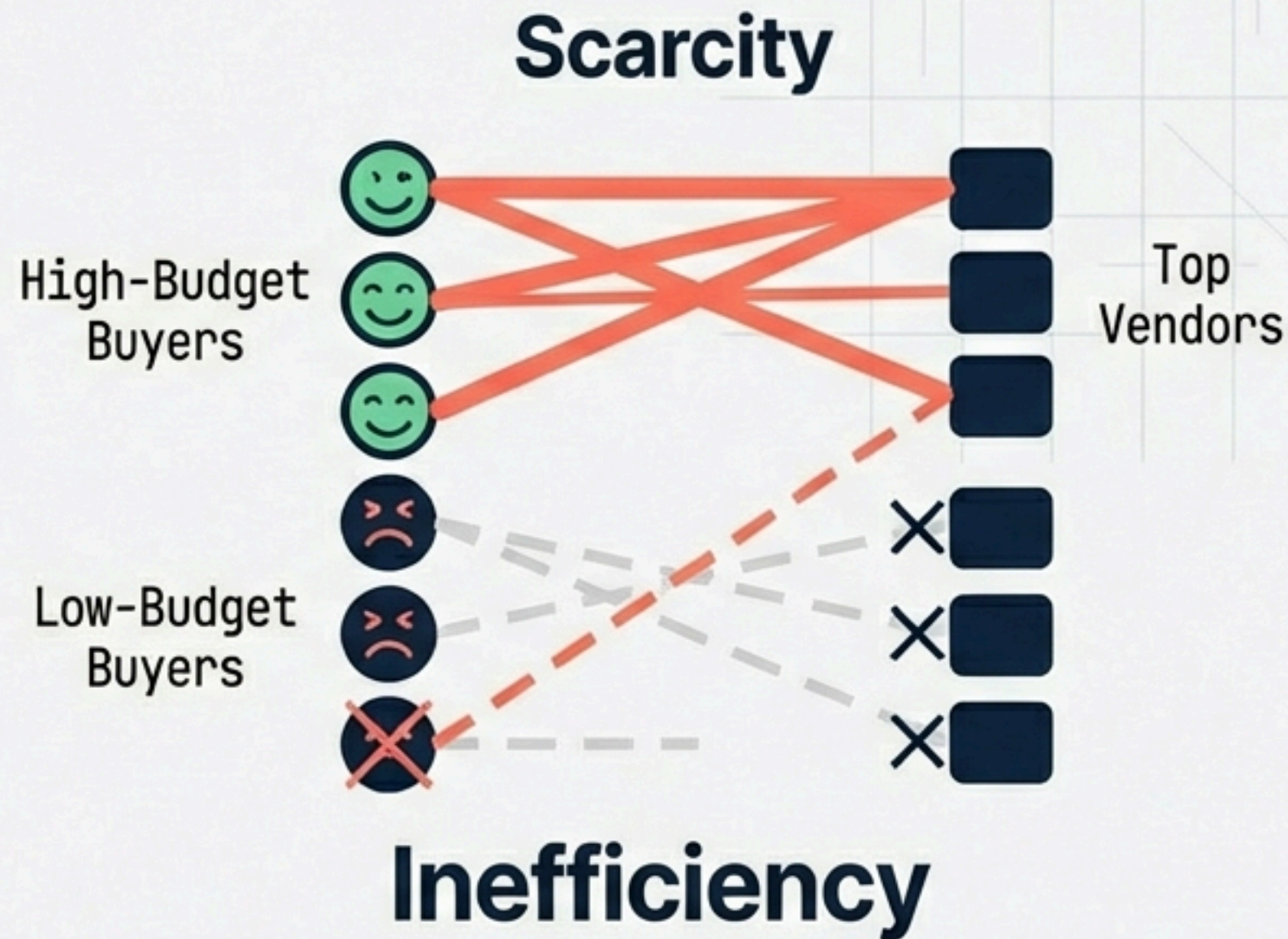
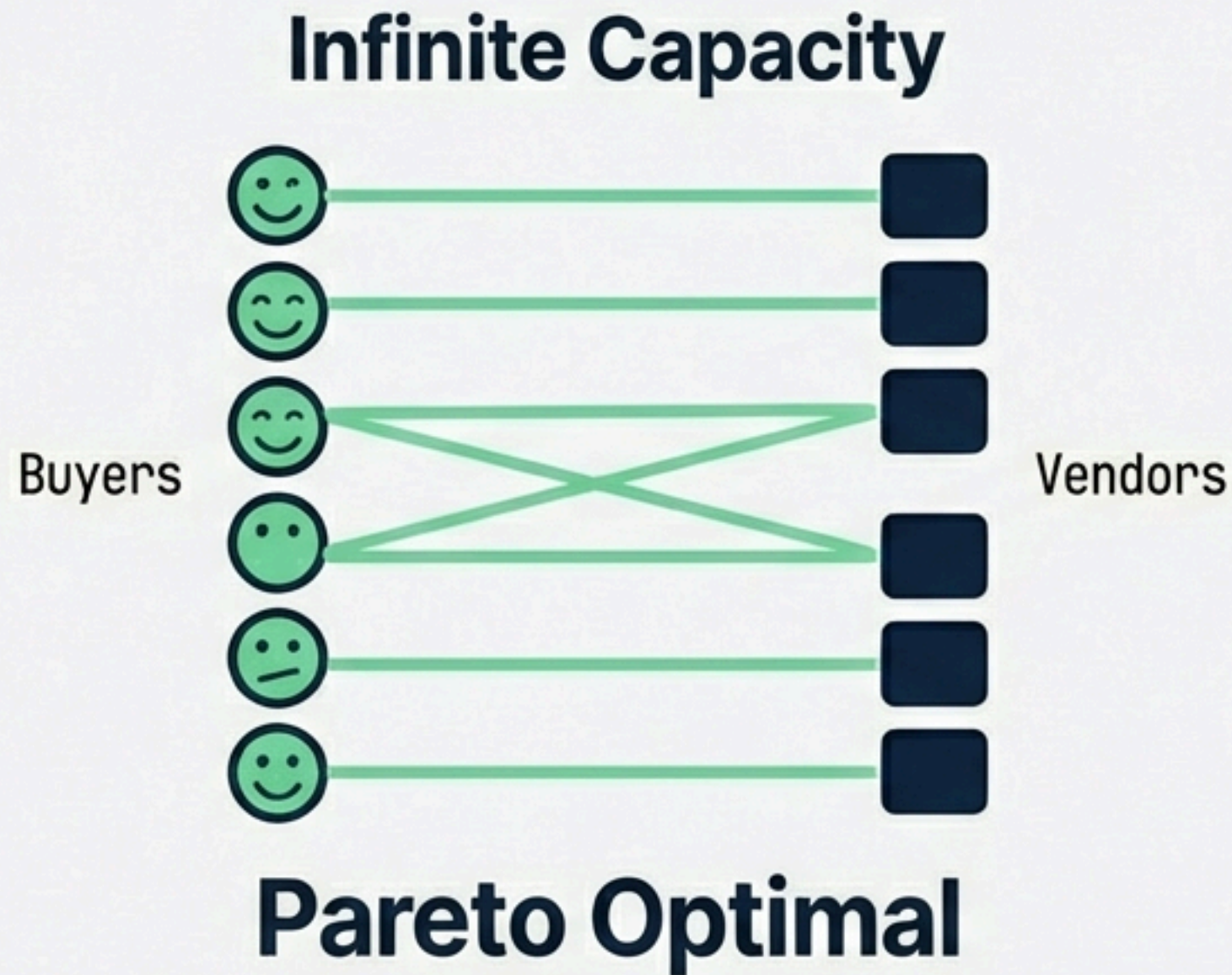
Providers (Quality Q, Capacity K)



$$\text{Utility} = (\text{Quality} \times 10) + (\text{Budget} - \text{Price})$$

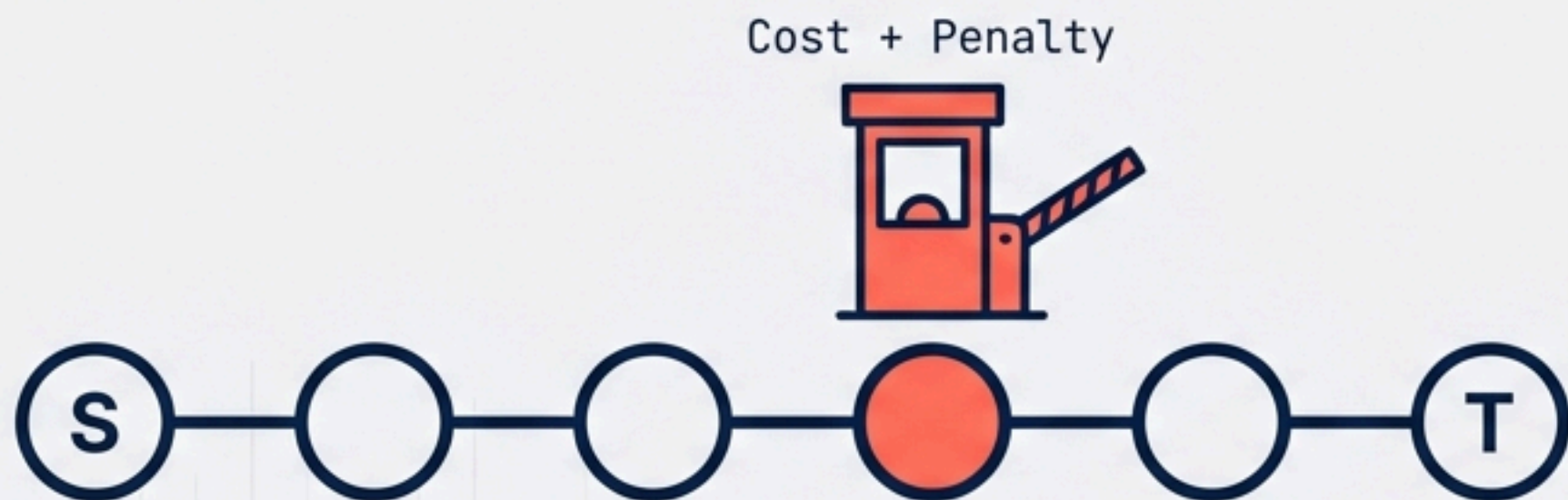
Goal: Maximize Social Welfare

Efficiency vs. Crowding Out



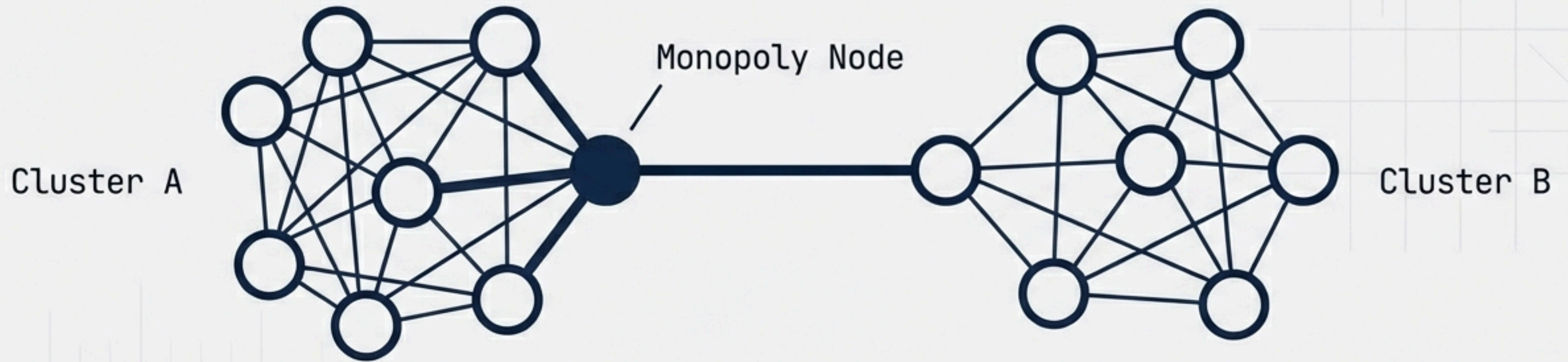
Key Insight: Infinite Supply: Local Greedy -> Global Optimum
Scarcity: Wealthier buyers crowd out others

Task 4: VCG Path Auction



- **Goal:** Procure secure path $s \rightarrow t$
- **Mechanism:** VCG (Truthful)
- **Cost Function:** Bid + Insecurity Penalty (Externality)

The VCG Payment Rule



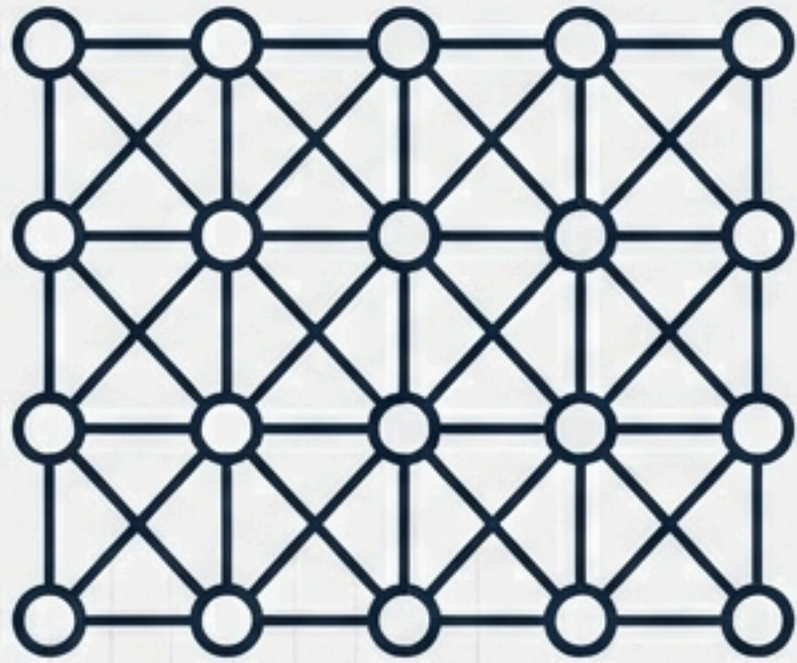
Payment = Externality

$$P_i = SP(G \text{ without } i) - (SP(G) - w_i)$$

Bridge nodes extract maximum payments.

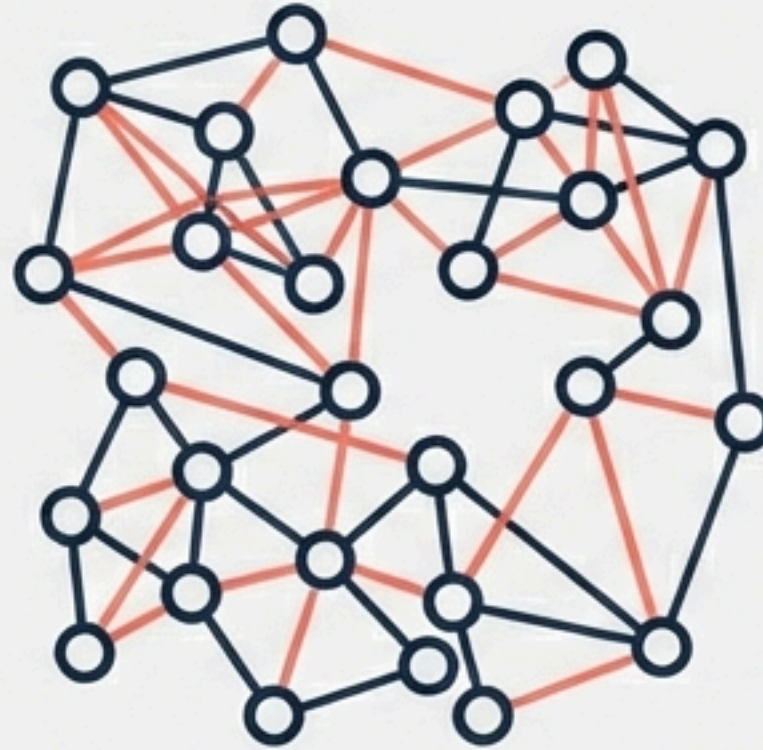
Topology Dictates Destiny

Regular Graph



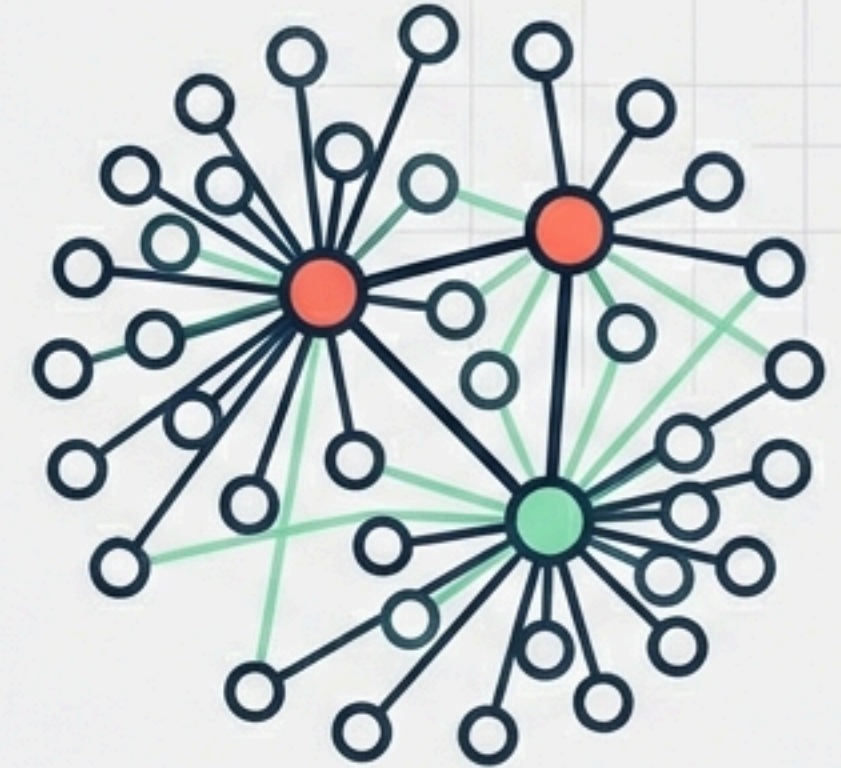
Distributed
Responsibility

Erdős-Rényi



Local Clusters

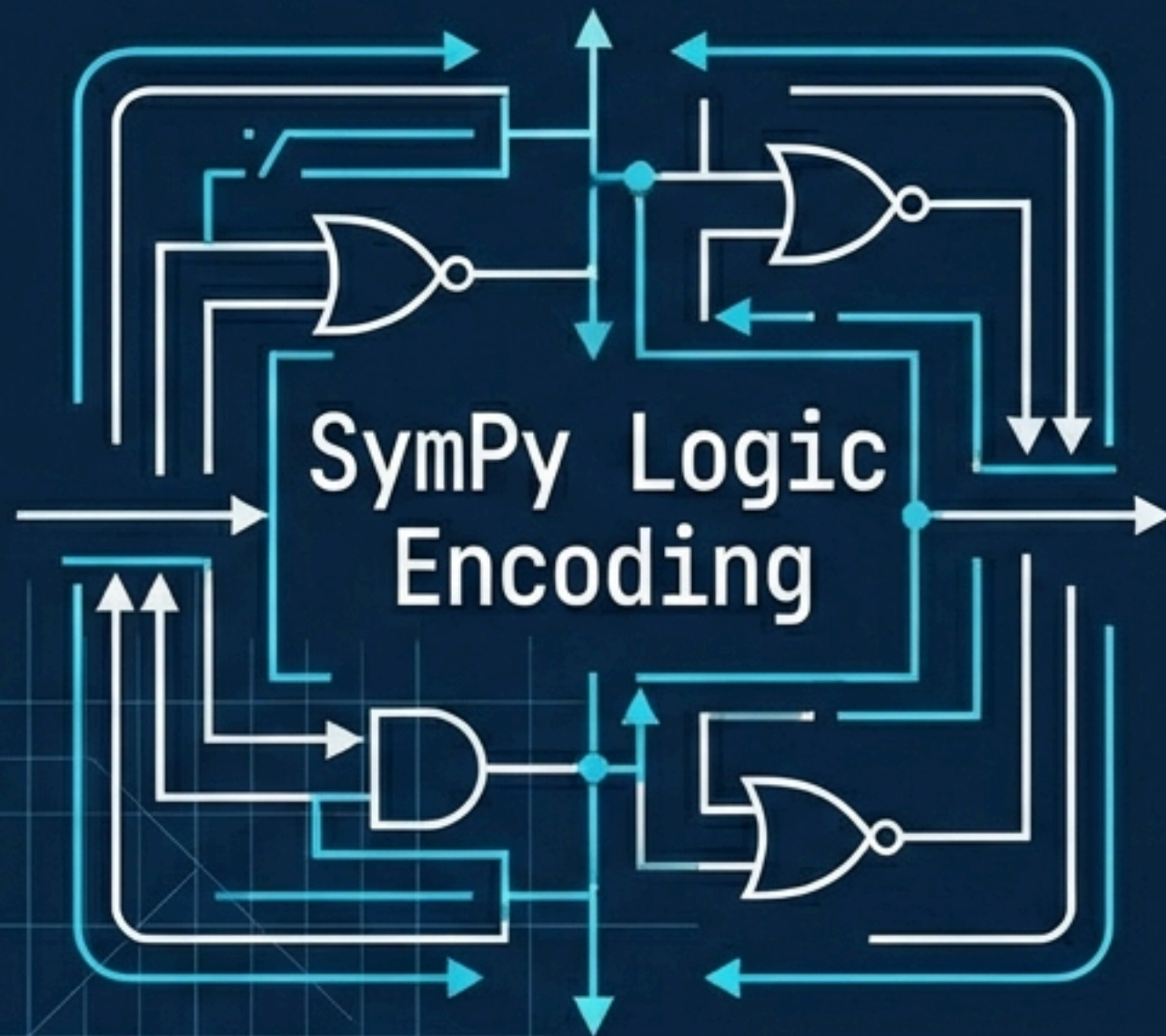
Barabasi-Albert



Critical Hubs
(Public Goods)

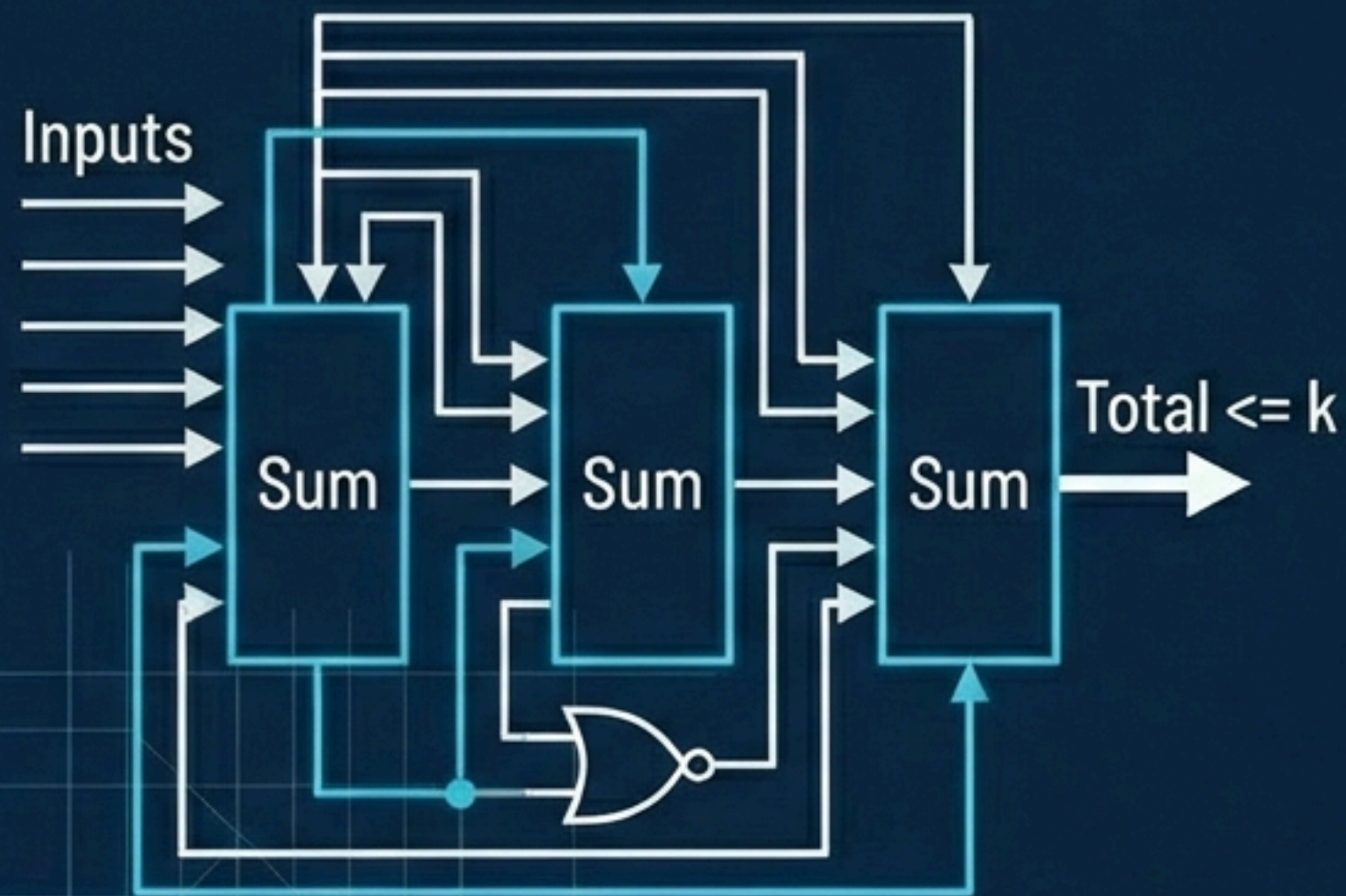
Network shape determines cost and market power.

Bonus: Exact Logic via SAT



- **Objective:** Algorithmic exploration beyond Game Theory.
- **Approach:** Boolean Satisfiability (SAT).
- **Encoding:** $(x_u \text{ OR } x_v)$ for every edge.

SAT Logic: The Cardinality Constraint



- **Challenge:** Encoding "At most k " without arithmetic.
- **Solution:** Recursive Sequential Counter Circuit.
- **Optimization:** Binary Search on k .

Conclusion & Key Insights



- **Integration:** From Selfish Agents to Boolean Logic.
- **Result:** Game Theory heuristics approached SAT Ground Truth.
- **Takeaway:** Topology is the primary determinant of security cost.



Thank You

Questions?

